

Math708 – Foundations of Computational Mathematics I
Fall 2011

Midterm Exam 2

You must show all of your work to get a credit for a correct answer.

1. (20 points) Let $\mathcal{I}_n(f) = (b-a) \sum_{k=0}^n \sigma_k f(x_k)$ be the Lagrange interpolation quadrature formula for the approximation of $\int_a^b f(x) dx$. If $a = -1$, $b = 1$, and

$$\mathcal{I}_2(f) = \frac{1}{3} \left(4f\left(-\frac{1}{2}\right) - 2f(0) + 4f\left(\frac{1}{2}\right) \right)$$

What is the degree of exactness of this $\mathcal{I}_2(f)$?

2. (20 points) Compute the coefficients and weights $\{(\lambda_j, \sigma_j)\}_{j=1}^n$ of the Gaussian quadrature formulas $\mathcal{I}_n(f) = \sum_{j=1}^n \sigma_j f(\lambda_j)$ for the approximation of $\int_a^b f(x) \rho(x) dx$ where $\rho(x) = \sqrt{x}$, $a = 0$, $b = 1$, and $n = 2$.
3. (20 points) Consider solving the the initial value problem $y' = f(t, y)$, $y(t_0) = y_0$ using the one-step method

$$u_{n+1} = u_n + h[(1 - \theta)f(t_n, u_n) + \theta f(t_{n+1}, u_{n+1})].$$

where θ is constant parameter in $[0, 1]$. Show that this method is A-stable when $\theta \in [1/2, 1]$.

4. (20 points) Consider the multi-step scheme

$$u_{n+1} - \frac{1}{2}u_n - \frac{1}{2}u_{n-1} = \frac{h}{8}(3f_{n+1} + 8f_n + f_{n-1})$$

where $f_k \equiv f(t_k, u_k)$ for solving the initial value problem

$$y' = f(t, y), \quad y(t_0) = y_0.$$

Compute its order. Is it stable?

5. (20 points) Consider the Runge-Kutta method of the form

$$u_{n+1} = u_n + h(K_1 + 3K_2)/4$$

with $K_1 = f(t_n, u_n)$, $K_2 = f(t_n + \frac{2}{3}h, u_n + \frac{2}{3}hK_1)$ for solving the initial value problem

$$y' = f(t, y), \quad y(t_0) = y_0.$$

- (a) Using Taylor's theorem to compute the order of consistency for the method.
- (b) Define what it means for $\eta = \lambda h$ to be in the region of absolute stability for the above scheme.